SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)
Accredited by NAAC with "A+" Grade.
Recognised as Scientific and Industrial Research Organisation
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation:R26	**I/IV-B.Tech. I -Semester**

MECHANICAL ENGINEERING

COURSE STRUCTURE

(With effect from 2026-27 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
A2610001	English for Technical Communication	**HS**	3	0	0	3	40	60	100
A2610002	Linear Algebra and Calculus	**BS**	3	0	0	3	40	60	100
A2610301	Physics for Mechanical Systems	**BS**	3	0	0	3	40	60	100
A2610007	Basic Electrical and Electronics Engineering	**ES**	3	0	0	3	40	60	100
A2610008	Design Thinking and Innovation	**ES**	0	1	3	2.5	50	50	100
A2610009	AI Tools and Applications	**ES**	0	1	3	2.5	50	50	100
A2610016	Workshop Practice	**ES**	0	0	3	1.5	50	50	100
A2610302	Physics for Mechanical Systems LAB	**BS**	0	0	3	1.5	50	50	100
A2610303	English for Technical Communications and MOODLES lab	**HS**	0	0	3	1.5	50	50	100
A2610017	Universal Human Values - II	**MC**	2	0	0	0	50	-	50
		TOTAL	**14**	**2**	**15**	**21.5**	**460**	**490**	**950**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610001	HS	3	--	--	3	40	60	3 hrs

ENGLISH FOR TECHNICAL COMMUNICATION

(Common to all Branches)

Course Objectives:

1.	Enable students identify the context and thematic information from social, academic, and workplace dialogues and texts.
2.	Develop students' ability to use appropriate grammatical structures and vocabulary.
3.	Equip students with contextual vocabulary for writing.
4.	Familiarize students with reading strategies to comprehend and interpret texts.
5.	Develop students' ability to demonstrate appropriate social media etiquette.

Course Outcomes

Sl. No	Outcome	Knowledge Level
1.	Identify the thematic information from transactional dialogues and texts to construct coherent written and spoken discourses.	K2
2.	Apply appropriate grammatical structures and vocabulary to summarize texts on the related themes.	K3
3.	Apply contextual vocabulary to paraphrase texts and write letters.	K3
4.	Develop reading strategies to comprehend and interpret various texts for a global understanding.	K3
5.	Construct book/film reviews and demonstrate appropriate social media etiquette in writing, employing sentence synthesis techniques.	K3

SYLLABUS

UNIT-I (10Hrs)	**Effective Communication: Exploring Channels and Methods** **Vocabulary:** Roots, Prefixes, and Suffixes; Homonyms **Grammar:** Articles, Prepositions, Tenses. **Listening:** Listening for main ideas and supporting details, Note-taking **Speaking:** Introducing self and others, Describing people and objects, JAM **Reading:** Definition and Importance, Skills and Sub-Skills of Reading **Writing:** Cohesion and Coherence, Paragraph Writing
UNIT-II (10 Hrs)	**Eco-Consciousness: Negotiating for a Sustainable Future** **Vocabulary:** Contractions, Phrasal Verbs, Idioms & Phrases **Grammar:** Subject-Verb Agreement, Conditional Sentences **Listening:** Listening to various texts and comprehension **Speaking:** Role-plays, Extempore **Reading:** Reading for Inference, Point of View, Tone and Perspective

	Writing: Note-making, Summarizing, Paraphrasing (at basic level)
UNIT-III (10 Hrs)	**Disruptive Technologies: Ethical Implications and Future Prospects** **Vocabulary:** One-Word Substitutes, Commonly Confused Words **Grammar:** Degrees of Comparison, Active and Passive Voice, Question Tags **Listening:** Listening to stories, news, documentaries, and discussions **Speaking:** Debate, Group Discussions **Reading:** Reading the extracts/articles/ essays of the newspapers, journals, and literary texts. **Writing:** Paraphrasing (at advanced level), Letter Writing
UNIT-IV (10 Hrs)	**Entrepreneurship and Start-Up Culture: Emerging Trends** **Vocabulary:** Business/ Profession/ Career/ Job-related Vocabulary **Grammar:** Sentence Synthesis, Simple, Compound and Complex Sentences **Listening:** Listening to debates and discussions **Speaking:** PowerPoint Presentations by Individuals and Teams **Reading:** Reading general texts for global understanding **Writing:** Email Writing, Data Interpretation, Report Writing
UNIT-V (10 Hrs)	**Ethical Leadership: Integrating Social Responsibility and Core Values** **Vocabulary:** Advanced Vocabulary for GRE/TOEFL/ IELTS/CAT **Grammar:** Sentence Synthesis, Para Jumbles **Listening:** Listening to talks/speeches by eminent personalities **Speaking:** Team Discussions and Presentations **Reading:** Reading specific texts for global comprehension **Writing:** Book/Film Reviews, Social Media Etiquette
Textbook:	
1.	Chekuri, T., & Perumalla, R. K. (2026). NextGen English for engineers: A course book for technical English communication. Cengage.
Reference Books:	
1.	Technical Communication (2nd Ed.) by M. Raman & S. Sharma. Oxford University Press, 2012.
2.	Effective Technical Communication by M.A. Rizvi, Tata McGraw-Hill, 2005.
3.	English for Engineers and Technologists, Orient Blackswan 2010 (Combined edition, Vol.1 &2).
4.	The Oxford Guide to Writing and Speaking by John Seely, Oxford University Press, 2004.
e-Resources	
1.	eGyanKosh. (n.d.). National digital repository. https://egyankosh.ac.in/
2.	Read Theory. (n.d). Adaptive reading comprehension practice. https://readtheory.org/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610002	BS	3	--	--	3	40	60	3 Hrs.

LINEAR ALGEBRA & CALCULUS

(Common to All Branches of Engineering)

Course Objectives: Students are expected to learn

1.	Concepts of linear algebra and methods of solution of linear simultaneous algebraicequations.
2.	Eigen values, Eigen vectors and quadratic forms.
3.	Problems and applications of Mean value theorems
4.	Application of partial differentiation for determining maxima/ minima of functions.
5.	Concepts of double, triple integrals and its applications.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Solve** a system of linear equations by using direct and indirect methods.	K3
2.	**Determine** Eigen values and Eigen vectors, execute matrix Diagonalization and reduce Q.F.'s to canonical forms.	K3
3.	**Apply** mean value theorems to analyze function behaviour and solve real life problems.	K3
4.	**Apply** the concepts of partial differentiation in various engineering applications.	K3
5.	**Apply** the concepts of multiple integrals to calculate areas, volumes of geometrical regions and in engineering applications.	K3

SYLLABUS

UNIT-I (10Hrs)	**Linear System of Equations:** Rank of a matrix by echelon form, normal form. Inverse of a matrix by Gauss-Jordan method. Consistency of linear equations: Homogeneous and Non-Homogeneous systems. Gauss elimination method, Jacobi and Gauss Seidel Iteration Methods.
UNIT-II (10 Hrs)	**Eigen Values, Eigen Vectors and Quadratic Forms:** Eigen values and Eigen vectors of a matrix, their properties, Cayley-Hamilton Theorem (without proof), Inverse and powers of a matrix by Cayley – Hamilton theorem, Diagonalization of a matrix, Quadratic forms and Nature of the Quadratic Forms, Reduction of a Quadratic form to canonical form by Orthogonal Transformation and Singular Value Decomposition.
UNIT-III (10 Hrs)	**Single Variable Calculus:** Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem, Geometrical interpretation of these theorems, Cauchy's mean value theorem, Taylor's and Maclaurin's theorems (without proof) with Lagrange's form of remainder, Taylor's and Maclaurin series.

UNIT-IV (10 Hrs)	**Partial differentiation and Applications:** Functions of several variables, Continuity and Differentiability, Partial derivatives, Euler's theorem, Total derivative, chain rule, Taylor's and Maclaurin's series expansion of function of two variables, Jacobians, Functional dependence, Maxima and Minima of function of two variables without constraints and with constraints - Method of Lagrange multipliers.
UNIT-V (10 Hrs)	**Multiple Integrals:** Double integrals, change of order of integration, triple integrals, change of variables from Cartesian to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals).

Textbooks:	
1.	HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition
2.	AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition.
Reference Books:	
1.	ThomasCalculus,GeorgeB.Thomas,MauriceD.WeirandJoelHass,PearsonPublishers,2018, 14thEdition.
2.	Higher Engineering Mathematics, B.V.Ramana, McGraw HillEducation,2017.
3.	Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4.	Advanced Engineering Mathematics, Micheael Greenberg, Pearson publishers, 9th edition.
5.	Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications,2014, Third Edition (Reprint 2021).
6.	Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty Two EditionReprint 2022).
7.	Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015.

e-Resources	
1.	https://en.wikipedia.org/wiki/System_of_linear_equations
2.	https://en.wikipedia.org/wiki/Eigenvalues_and_eigenvectors
3.	https://www.math.hmc.edu/calculus/tutorials/eigenstuff/
4.	https://en.wikipedia.org/wiki/Quadratic_form
5.	https://en.wikipedia.org/wiki/Calculus
6.	https://en.wikipedia.org/wiki/Partial_derivative
7.	https://www.whitman.edu/mathematics/calculus_online/section14.03.html
8.	https://en.wikipedia.org/wiki/Multiple_integral
9.	http://tutorial.math.lamar.edu/Classes/CalcIII/MultipleIntegralsIntro.aspx

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610301	BS	3	--	--	3	40	60	3 Hrs.

PHYSICS FOR MECHANICAL SYSTEMS

(For ME)

Course Objectives:

1.	To provide students with a strong foundation in wave optics concepts such as interference, diffraction and polarization along with their applications in optical measurements.
2.	To introduce the fundamentals of crystallography and X-ray diffraction methods for understanding crystal structures and material characterization.
3.	To equip students, the basic principles of quantum mechanics and quantum computing, including quantum states, tunneling, superposition and secure communication.
4.	To familiarize students with oscillations, vibration control systems, lasers and precision engineering applications in modern industries.
5.	To impart knowledge on the principles of ultrasonics and non-destructive testing methods for defect detection and quality inspection.

Course Outcomes: After successful completion of this course, the student will be able to:

S.No.	Outcome	Knowledge Level
1.	Examine the principles of interference, diffraction and polarization and utilize optical analysis techniques in engineering applications.	K3
2.	Analyze crystal structures, diffraction patterns and material characterization using X-ray diffraction methods.	K4
3.	Interpret the concepts of quantum mechanics and quantum computing and apply for secure communication systems.	K3
4.	Analyze oscillatory systems, vibration control mechanisms and laser-based metrology techniques.	K4
5.	Evaluate ultrasonic testing and non-destructive testing methods for defect identification and quality assessment.	K4

SYLLABUS

UNIT-I (10Hrs)	**APPLIED WAVE OPTICS** **Interference:** Superposition Principle – Coherence – Types of Interference – Thin-film interference (reflection geometry) – Newton's rings – Determination of wavelength and refractive index. **Diffraction and Optical Surveying:** Fresnel and Fraunhofer diffraction – Fraunhofer's diffraction due to single slit(quantitative), double slit & N-slits (qualitative) – Diffraction Grating and grating spectrum – Dispersive and Resolving power of a grating (qualitative). **Polarization:** Types of polarization – Nicol's prism – Half-wave and Quarter-wave plates.
UNIT-II	**CRYSTALLOGRAPHY AND MATERIAL CHARACTERIZATION**

(9 Hrs)	**Crystallography:** Space lattice – Basis – Unit cell – Lattice parameters – Crystal systems and Bravais lattices, Co-ordination number and packing fraction of SC, BCC and FCC structures. **Material Characterization:** Miller indices – Inter-planar spacing between successive (h k l) planes – Bragg's law – X-ray Diffractometer – Crystal structure determination by Laue's method.
UNIT-III (10 Hrs)	**INTRODUCTION TO QUANTUM MECHANICS & QUANTUM COMPUTING** **Quantum Mechanics:** de Broglie concept of matter waves – Heisenberg's uncertainty principle – Wave function – Postulates of quantum mechanics – Schrödinger's time dependent and time-independent wave equations -–Particle in a one-dimensional potential well – Quantum tunneling (basic idea). **Quantum Computing:** Introduction to quantum computing – Classical bits vs qubits – Superposition and entanglement – Basic quantum operations – Hadamard gate – Pauli-X gate – Idea of quantum measurement.
UNIT-IV (10 Hrs)	**OSCILLATIONS AND LASER-BASED METROLOGY** **Oscillations:** Oscillations of springs – Series and parallel combination of springs (quantitative) – Stiffness factor and its significance – Damping and resonance: over-, under- and critical damping (qualitative) – Forced oscillations and resonance – Sharpness of resonance – Quality factor (Q). **Lasers and Precision Engineering:** Principles of Absorption, Spontaneous and Stimulated Emission, Population Inversion. Three-level and Four-level laser systems, Construction and working of Ruby and He-Ne Laser, Applications (qualitative): Role of high-power lasers in CNC Laser Cutting and Welding.
UNIT-V (9 Hrs)	**ULTRASONICS AND NON-DESTRUCTIVE TESTING** **Ultrasonics:** Introduction – Characteristics of ultrasonic waves – Production of ultrasonic waves by magnetostriction and piezoelectric methods – Detection - Acoustic grating – Applications. **Non-Destructive Testing (NDT):** Introduction - pulse echo system through transmission and reflection modes – Applications.

Textbooks:	
1.	Avadhanulu, M.N., Kshirsagar, P.G., & Murthy, T.V.S.A., A textbook of Engineering Physics, S. Chand Publishing (2019)
2.	P.K. Palaniswamy, Engineering Physics, SciTech Publications (2019)
Reference Books:	
1.	Pandey, B.K., & Chaturvedi, S., Engineering Physics, Cengage Learning (2021)
2.	Bhattacharya, D.K., & Tandon, P., Engineering physics, Oxford University Press (2015)
3.	Srinivasan, M. R., Engineering Physics, New Age International (2009)

4.	Nielsen & Chuang, Quantum Computation & Quantum Information, Cambridge (2010)
e-Resources	
1.	NPTEL. (n.d.). National Programme on Technology Enhanced Learning. https://nptel.ac.in
2.	MIT OpenCourseWare. (n.d.). Physics courses. https://ocw.mit.edu
3.	Georgia State University. (n.d.). HyperPhysics. http://hyperphysics.phy-astr.gsu.edu
4.	PhET Interactive Simulations. (n.d.). University of Colorado Boulder. https://phet.colorado.edu
5.	American Concrete Institute. (2021). ACI 228.2R: Nondestructive test methods for evaluation of concrete in structures. https://www.concrete.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610007	ES	3	--	--	3	40	60	3 Hrs

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

(Common to AIDS, IT & ME)

Course Objectives: Students will learn about

1.	The basic principles of DC & AC Circuit analysis.
2.	The fundamentals of Electric power generation and measuring instruments.
3.	The Electrical Motors for Energy conversion and Electrical Safety.
4.	The fundamentals of semiconductor devices and their applications and basic electronic circuits.
5.	The fundamentals of Digital systems.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Apply the circuit laws to find responses in DC and AC Circuits.	**K3**
2.	Illustrate the components & working of power generating plants and measuring instruments.	**K3**
3.	Illustrate the working of various electric motors and electric safety devices.	**K3**
4.	Apply the principles of Diodes & BJTs to understand the working of rectifiers and voltage regulators.	**K3**
5.	Apply the basic principles of digital logic to understand the working of adders, flip-flops, registers and counters.	**K3**

SYLLABUS

UNIT-I (10Hrs)	**Direct Current (DC) & Alternating Current (AC) Circuits:** **DC Circuits:** Electrical circuit elements (R, L and C), Ohm's Law, Kirchoff's laws, series-parallel resistive circuits, Simple numerical problems with Voltage Sources. **AC Circuits:** A.C. Fundamentals - time period, frequency, amplitude, phase, phase difference, Sinusoidal voltages and currents, average value, RMS value of sinusoidal waveforms, Phasor representation of Voltages and currents, Impedance of R, L & C elements, average power & power factor of Series RL, RC and RLC circuits - Simple Numerical problems.
UNIT-II (10 Hrs)	**Electric Power Generation and Measuring instruments** Construction and principle of 3 – phase Alternator, Transformer principle, Major sources of electricity generation: schematics of conventional power plants (Thermal and Hydro), Non-conventional sources (solar and wind). Measuring Instruments: Digital Voltmeter (DVM), Digital Multimeter, Cathode Ray Oscilloscope (CRO) and Single-phase Digital Energy meter.

UNIT-III (10 Hrs)	**Electrical Energy Consumption and Safety Measures:** Major Electrical Loads, DC motor - Construction and Working principle, Torque equation, AC motor - Working principle of 3-phase Induction motor, slip - Other electrical machines: Stepper motor, BLDC Motor. **Electrical Safety:** Electric Shock, Safety Precautions to avoid shock, Earthing and its types, Protective devices: Fuse, Miniature circuit breaker (MCB), Earth leakage circuit breaker (ELCB) and Residual Current Circuit Breaker (RCCB).
UNIT-IV (10 Hrs)	**SEMICONDUCTOR DEVICES AND CIRCUITS** Introduction – Operation and Characteristics of PN Junction Diode, Rectifiers - working and analysis of a Half wave and full wave bridge rectifiers, capacitor filter (no analysis). Zener Diode and its Characteristics, Working of Zener voltage regulator. Principle & Operation of Bipolar Junction Transistor - Transistor as a switch and amplifier (CE configuration).
UNIT-V (10 Hrs)	**DIGITAL LOGIC FUNDAMENTALS** Overview of Number Systems – Binary, Hexa-decimal. Boolean Algebra - Basic Theorems - Truth Tables and Functionality of Logic Gates – NOT, OR, AND, NOR, NAND, XOR and XNOR. Simple combinational circuits–Half and Full Adders. Introduction to sequential circuits - Clocked S-R and J-K Flip-flops, Simple examples of two-bit Registers and Counters.

Textbooks:	
1.	Principles of Electrical and Electronics Engineering, V.K. Mehtha, S. Chand Technical Publishers, 2020
2.	Basic Electrical Engineering, Ritu SahDev, Khanna Publishers,2018, First Edition
3.	R. L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education,2021.
4.	Sanjeev Gupta & Santhosh Gupta, Electronic Devices & Circuit, Dhanpat Rai Publications,2010
Reference Books:	
1.	Non-conventional Energy sources by G.D Rai, Khanna Publishers, 2009, Third Edition
2.	Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, Mc Graw Hill, 2019, Fourth Edition
3.	Principles of Power Systems, V.K. Mehtha, S.Chand Technical Publishers, 2020
4.	Principles of Electrical and Electronics Engineering, V.K. Mehtha, S.Chand Technical Publishers, 2020
5.	R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009
6.	R. S. Sedha, A Textbook of Electronic Devices and Circuits, S. Chand & Co, 2010.
e-Resources	
1.	**https://nptel.ac.in/courses/108105053**

2.	**https://nptel.ac.in/courses/108108076**
3.	**https://nptel.ac.in/courses/108105112**
4.	**https://nptel.ac.in/courses/108101091**
5.	**https://nptel.ac.in/courses/108105113**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610008	**ES**	**--**	**1**	**3**	**2.5**	**50**	**50**	**3 Hrs.**

DESIGN THINKING AND INNOVATION

(Common to AIML, CIC, CIVIL, CSE, CSBS, ECE, EEE, MECH)

Course Objectives: At the end of the course, the teacher should be able to:

1	Introduce students to the principles, mindset and methodologies of **Design Thinking and human-centered problem solving**.
2	Develop students' ability to **observe, empathize, analyze user needs and formulate well-defined engineering and societal problems**.
3	Develop students' **creative and innovative thinking abilities** through systematic ideation, concept generation and evaluation techniques.
4	Enable students to **develop, prototype, test and iteratively improve solutions** using user feedback and appropriate physical/digital tools.
5	Develop students' ability to identify innovation opportunities, create value propositions, understand basic IPR and communicate innovative solutions effectively.

Course Outcomes: After successful completion of the course, student will be able to

S.No	Outcome	Knowledge Level
1	Explain the principles, processes and mindset of Design Thinking for engineering problem-solving and innovation	K2
2	Apply empathy, observation and problem-framing techniques to identify user needs and define engineering problems	K3
3	Apply creative and systematic ideation techniques to generate and select innovative solution concepts	K3
4	Analyze solution concepts and prototypes using user feedback and appropriate design criteria for iterative improvement	K4
5	Analyze the innovation potential of solutions considering value proposition, feasibility, sustainability, societal impact and basic IPR aspects	K4

1	**UNIT I — Design Thinking Foundations & Empathy** Introduction to Design Thinking ; Design thinking mindset and principles ; Design thinking vs. conventional engineering problem-solving ; Human-centered design ; Five-stage design thinking process ; Creativity, curiosity and observation ; Understanding users and stakeholders ; Empathy and user research ; Interviews, observation and contextual inquiry ; Empathy mapping ; Personas and user journeys ; Identifying explicit and latent/unarticulated needs ; Ethical considerations in user research **Practical Activities** • Observation of a real-life problem

	• Conducting user interviews • Preparing an empathy map • Developing a persona • Preparing a user journey map **Unit Outcome** Students will be able to **understand users, identify their needs and translate observations into meaningful design opportunities**
2	**UNIT II — Problem Definition, Analysis & Opportunity Identification** From observations to insights ; Data synthesis and affinity mapping ; Identifying patterns and pain points ; Problem statements ; Point-of-View (POV) statements ; "How Might We" (HMW) questions ; Stakeholder mapping ; 5 Whys and root-cause analysis ; Problem prioritization ; Need–Opportunity mapping ; Problem validation ; Design constraints and specifications ; Social, environmental and sustainability considerations ; Identifying opportunities for innovation **Practical Activities** • Affinity mapping of interview data • Develop three alternative problem statements • Prepare POV and HMW statements • Stakeholder map • Root-cause analysis • Select and validate one problem/opportunity **Unit Outcome** Students will be able to **analyze user research, identify root causes, frame well-defined problems and recognize opportunities for innovative solutions**.
3	**UNIT III — Creativity, Ideation & Concept Development** Creativity and innovation mindset; Divergent and convergent thinking ; Brainstorming and brainwriting ; SCAMPER ; Mind mapping ; Reverse brainstorming ; Analogical thinking ; Biomimicry ; Lateral thinking ; Forced connections ; Morphological analysis ; TRIZ — basic principles ; Individual and collaborative ideation ; AI-assisted ideation: opportunities and limitations ; Idea clustering and screening ; Impact–feasibility matrix ; Concept selection ; Design criteria and decision matrix **Practical Activities** • Generate 20–30 ideas for a selected problem • SCAMPER exercise • Brainwriting exercise • Biomimicry-based ideation • AI-assisted idea generation and critical evaluation • Prepare an idea-selection matrix • Select the top 2–3 concepts **Unit Outcome** Students will be able to **generate diverse solution concepts using creative techniques and systematically select promising ideas based on defined criteria**.

4	**UNIT IV — Prototyping, Experimentation & Validation** Concept development ; Prototype mindset ; Types of prototypes: Paper prototype, Physical prototype, Digital prototype, Storyboard, Role-play, Service prototype ; Low-fidelity and high-fidelity prototypes ; Rapid prototyping ; "Fail fast, learn fast" ; Minimum Viable Prototype (MVP) ; Design experimentation ; Prototype evaluation ; User testing ; Think-aloud method ; Feedback collection ; Usability and functionality testing; Iterative design ; Design critique ; Documentation of iterations ; Use of CAD, simulation and AI-assisted visualization tools **Practical Activities** • Develop a low-fidelity prototype • Prepare a storyboard • Build a physical/digital prototype • Conduct user testing • Collect structured feedback • Modify the prototype • Prepare an iteration log
5	**UNIT V — Innovation, Value Creation & Communication** **Innovation: meaning, characteristics and importance:** Invention vs. innovation ; Incremental, disruptive and radical innovation ; Product, process, service and social innovation ; Technology-driven vs. need-driven innovation ; Sustainable and frugal innovation ; Innovation opportunities and emerging technologies ; From prototype to innovation ; Desirability, feasibility and viability ; Value Proposition ; Value Proposition Canvas ; Introduction to Business Model Canvas ; Customer segments and stakeholder value; Innovation validation; Market/user validation ; Intellectual Property Rights — introduction; Patents, copyright, trademarks and design protection ; Responsible innovation ; Inclusive and sustainable innovation ; Innovation ecosystem and entrepreneurship ; Communicating innovative ideas ; Innovation pitch **Practical Activities** 1. Classify existing products/services as incremental, disruptive or radical innovations. 2. Identify three innovation opportunities from a campus/community problem. 3. Prepare a **Value Proposition Canvas**. 4. Validate the proposed innovation with potential users. 5. Prepare an introductory **Business Model Canvas**. 6. Identify possible IPR protection. Prepare and deliver a **3-minute Innovation Pitch**
Text Books:	
1	Chandan Adhikari, Prabir Kumar Das & Tanay Pramanik, Textbook of Design Thinking and Innovation, S. Chand Publishing, 2026.
2	Shalini Rahul Tiwari & Rohit Rajendra Swarup, Design Thinking: A Comprehensive Textbook, Wiley India, 2024

Reference Books:	
1	Soni, Pavan, Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving, Penguin Random House India.
2	Jauhari, Vinnie & Sudhanshu Bhushan, Innovation Management, Oxford University Press India.
Web References:	
1	Atal Innovation Mission, NITI Aayog – Innovation and Design Thinking Resources. https://aim.gov.in/
2	Institution's Innovation Council (IIC), Ministry of Education, Government of India, https://iic.mic.gov.in/
3	Startup India, Government of India – Innovation and Entrepreneurship Ecosystem, https://www.startupindia.gov.in/
4	National Innovation Foundation – India – Grassroots and Frugal Innovation, https://nif.org.in/
5	Stanford d.school – Design Thinking Bootleg and Design Thinking Resources, https://dschool.stanford.edu/resources/design-thinking-bootleg
6	IDEO.org – Human-Centered Design and Social Innovation Resources, https://www.ideo.org/
7	IDEO U – Design Thinking and Innovation Resources, https://www.ideou.com/pages/design-thinking
8	NPTEL – Online Courses on Design, Innovation, Creativity and Entrepreneurship, https://nptel.ac.in/
9	SWAYAM – Online Courses on Innovation, Entrepreneurship and Design Thinking, https://swayam.gov.in/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610009	ES	--	1	3	2.5	50	50	3Hrs

AI Tools & Applications

(Common to AIDS, CIVIL, CSBS, CSD, CSIT, EEE, IT, MECH)

Course Objectives: To familiarize students with the landscape of AI tools and their underlying working principles.

1	To develop competence in prompt engineering for effective interaction with generative AI systems
2	To enable students to apply AI tools for data analysis, visualization, and interpretation.
3	To train students in using AI-assisted tools for coding, writing, and productivity tasks.
4	To instill awareness of ethical, legal, and societal considerations in AI tool usage.

Course Outcomes:

S. No	Outcome	Knowledge Level
1	Illustrate the categories, capabilities, and limitations of major AI tools (text, image, code, data).	K3
2	Apply the principles of prompt engineering and demonstrate how generative AI models process instructions.	K3
3	Apply AI tools to solve first-year engineering problems in analysis, writing, and coding tasks.	K3
4	Analyze AI-generated outputs for accuracy, bias, and reliability before use.	K4
5	Evaluate the ethical, academic-integrity, and societal implications of AI tool use, justifying appropriate usage choices.	K4

SYLLABUS

1

Introduction to AI and the AI Tools Landscape What is AI, Machine Learning, and Generative AI — basic definitions and how they relate to one another. Core categories of AI tools: conversational/text, code assistants, data-analysis tools, image-generation tools, and writing/productivity tools. How large language models process and generate responses (conceptual, non-mathematical overview). Criteria for selecting the right AI tool for a given task — accuracy needs, cost, data sensitivity, and output format. Setting up accounts and workspaces; understanding free-tier versus paid-tier capabilities and limitations.

Widely Used AI Tools by Application (for classroom coverage):

Category	Example Tools	Typical Application
Conversational/General-Purpose AI	ChatGPT, Claude, Gemini, Perplexity	Writing, research, coding help, file/data analysis, cited web search
AI Coding Assistants	GitHub Copilot, Cursor, Claude Code	In - IDE code completion, AI Native editing, multi-file agentic coding

	Data Analysis Tools	ChatGPT(Advanced Data Analysis), Microsoft Copilotin Excel, Julius AI	Upload a dataset, generate statistics/charts, spreadsheet Formulas via natural language
	Image Generation	Mid journey, DALL·E 3, Stable Diffusion/Flux	Illustrations, concept art, design mockups
	Writing & Productivity	Grammarly, Gamma, Fathom, Notion AI	Grammar/style checking, slide generation, meeting summaries, task automation
	Audio, Video & Presentation (optional)	Eleven Labs, Runway/Sora, Synthesia	Text-to-speech, AI video generation, script-to-video presentations

Teacher Practical Examples (for classroom demonstration):

1. Live comparison of the same general engineering query (e.g., "explain the engineering design process") answered by two different AI chatbots, highlighting differences in response style and accuracy.

2. Walkthrough of an AI tool's interface — settings, conversation history, file upload, and model selection — using a common platform.

Student Exercise Problems (for practice):

1. Create accounts on two AI platforms and compare their responses to a simple engineering question (e.g., "explain the difference between analysis and design").

2. Prepare a one-page comparison chart of five AI tools by category and primary use case.

3. Write a short reflection identifying one task in your daily academic life where an AI tool could help.

2

Prompt Engineering and Generative AI Tools

Anatomy of a good prompt — context, clear instruction, desired format, and constraints.

Zero-shot, few-shot, and chain-of-thought prompting techniques and when to use each.

Role-based and context-setting prompts (e.g., asking the AI to respond as a reviewer, tutor, or domain expert).

Iterative prompt refinement — diagnosing a poor AI response and rewriting the prompt to fix it.

Using generative AI for brainstorming, summarizing, and drafting technical content.

Teacher Practical Examples (for classroom demonstration):

1. Refine a vague prompt (e.g., "explain project scheduling") into a precise one step-by-step, showing output improvement at each stage.

2. Demonstrate few-shot prompting by providing examples to guide the AI toward a specific output format (e.g., a formatted comparison table).

Student Exercise Problems (for practice):

1. Write three versions of a prompt (basic, detailed, role-based) for the same task and compare outputs.

2. Use an AI tool to summarize a one-page technical article and verify the summary against the original.

3. Draft a chain-of-thought prompt to solve a multi-step numerical/logical problem (e.g., a unit-

	conversion or estimation problem) and check the reasoning.
3	**AI Tools for Data Analysis and Visualization** AI-assisted spreadsheet and dataset analysis using natural-language commands. Natural-language querying of datasets to extract summary statistics and insights. AI tools for generating charts, graphs, and visual summaries from tabular data. Interpreting AI-generated statistical summaries correctly and recognizing their limitations. Identifying and correcting errors or misinterpretations in AI-generated data analysis. **Teacher Practical Examples (for classroom demonstration):** **1.** Upload a small dataset (e.g., class attendance or survey data) to an AI tool and generate a summary-statistics report and chart using natural-language commands. **2.** Show an example where an AI tool misinterprets data (e.g., treats a numeric code as text), and demonstrate how to verify and correct it. **Student Exercise Problems (for practice):** **1.** Use an AI tool to clean and summarize a small sample dataset (e.g., class attendance or marks data). **2.** Generate three different chart types from the same dataset using AI prompts and justify which is most appropriate. **3.** Cross-check one AI-generated statistical claim manually and report any discrepancy found.
4	**AI Tools for Coding, Writing, and Productivity** AI coding assistants — code generation, explanation, and debugging support for simple programs. AI writing assistants for reports, emails, and technical documentation. AI tools for scheduling, note-taking, and meeting/lecture summarization. AI-assisted presentation and slide-outline generation from a topic prompt. Integrating AI tools into an efficient day-to-day academic and engineering workflow. **Teacher Practical Examples (for classroom demonstration):** **1.** Use an AI coding assistant to generate a simple program (e.g., a temperature converter), then intentionally introduce a bug and have the AI debug it. **2.** Generate a presentation outline and speaker notes from a topic prompt using an AI writing/productivity tool. **Student Exercise Problems (for practice):** **1.** Use an AI coding tool to write and debug a simple program for a given problem statement. **2.** Draft a formal email or short report using an AI writing assistant, then manually edit it for accuracy and tone. **3.** Generate a slide outline for a 5-minute seminar topic using an AI tool and refine it.
5	**Ethics, Reliability, and Emerging Trends** Bias and hallucination in AI outputs, and techniques for detecting and fact-checking them. Academic integrity and appropriate citation of AI assistance in coursework and submissions. Data privacy and security considerations, including obligations under India's Digital Personal Data Protection Act, 2023 (DPDP Act), when uploading personal or institutional data. Copyright and intellectual property considerations in AI-generated text and image content. Emerging trends — multimodal AI and AI agents — and their implications for future

	engineering practice. **Teacher Practical Examples (for classroom demonstration):** **1.** Show a documented case of AI hallucination (a fabricated fact or citation) and demonstrate how to fact-check it. **2.** Walk through an example of properly citing AI assistance in an academic submission per institutional guidelines. **3.** Walk through a worked example of checking whether AI-generated text or image output raises a copyright/IP concern (e.g., close paraphrasing of a source). **Student Exercise Problems (for practice):** **1.** Identify and document a hallucination or bias instance produced by an AI tool on a factual query. **2**. Draft an "AI usage declaration" for a hypothetical assignment, citing tools and extent of use appropriately. **3.** Review the data-handling/privacy policy of one AI tool used in this course and identify one DPDP Act, 2023 compliance consideration relevant to student data.
Text Books:	
1	Marr, B. (2024). Generative AI in practice: 100+ amazing ways generative artificial intelligence is changing business and society. Wiley.
2	Mollick, E. (2024). Co-intelligence: Living and working with AI. Portfolio/Penguin.
Reference Books:	
1	Selwyn, N. (2022). Education and technology: Key issues and debates (3rd ed.). Bloomsbury.
2	Susskind, R., & Susskind, D. (2022). The future of the professions: How technology will transform the work of human experts (Updated ed.). Oxford University Press.
3	Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson.
Web Resources:	
1	Google. (n.d.). Google AI essentials. Grow with Google. https://grow.google/ai-essentials/
2	OpenAI. (n.d.). OpenAI documentation and guides. https://platform.openai.com/docs
3	Anthropic. (n.d.). Prompt engineering overview. Claude Docs. https://docs.claude.com/en/docs/build-with-claude/prompt- Engineering/overview
4	UNESCO. (2023). Guidance for generative AI in education and research. https://unesdoc.unesco.org/
5	AICTE. (2025). AICTE inputs on Artificial Intelligence — Year of AI initiative. https://www.aicte.gov.in/downloads/initiatives/AICTE%20inputs%20on%20Artificial%20Intelligence%20-compre hensive.pdf
6	Ministry of Electronics and Information Technology, Government of India. (2023). Digital Personal Data Protection Act, 2023. https://www.meity.gov.in/content/digital-personal-data-protection-act-2023

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610016	ES	--	--	3	1.5	50	50	3 Hrs

WORKSHOP PRACTICE

(Common for CE & ME)

Course Objectives:

1	Familiarize students with workshop tools, equipment, materials, and safety practices used in basic engineering activities.
2	Develop hands-on skills in wood working, sheet metal working, fitting, foundry, welding, and plumbing operations.
3	Provide practical experience in basic electrical wiring, soldering, and electrical circuit connections.
4	Introduce students to basic tinkering, sensors, actuators, microcontrollers, and IoT applications.
5	Develop students' practical problem-solving, fabrication, assembly, maintenance, and troubleshooting skills through hands-on activities.

Course Outcomes:

S.No	Outcome	Knowledge Level
1	Apply safety practices and identify appropriate workshop tools, equipment, materials, and measuring instruments for basic engineering activities	K3
2	Perform basic fabrication and manufacturing operations using carpentry, sheet metal, fitting, foundry, welding, and plumbing techniques.	K3
3	Apply workshop practices for the assembly, disassembly, maintenance, and repair of simple engineering components and systems.	K3
4	Perform basic electrical wiring, circuit connections, soldering, and testing using appropriate electrical practices.	K3
5	Develop simple sensors, actuator, microcontroller, and IoT-based applications through hands-on tinkering and experimentation.	K5

SYLLABUS

1	Demonstration: Safety practices and precautions to be observed in workshop.
2	Wood Working: Familiarity with different types of wood and tools used in wood working and making following joints. a) Half – Lap joint b) Mortise and Tenon joint c) Corner Dovetail joint or Bridle joint
3	Sheet Metal Working: Familiarity with different types of tools used in sheet metal working, Developments of following sheet metal job from GI sheets. a) Tapered tray b) Conical funnel c) Elbow pipe d) Brazing
4	Fitting: Familiarity with different types of tools used in fitting and do the following fitting exercises. a) V-fit b) Dovetail fit c) Semi-circular fit

5	Electrical Wiring: Familiarity with different types of basic electrical circuits and making the following connections. a) Parallel and series b) Two-way switch c) Godown lighting
6	Foundry Trade: Demonstration and practice on Moulding tools and processes, Preparation of Green Sand Moulds for given Patterns.
7	Welding Shop: Demonstration and practice on GTAW. Preparation of Lap joint and Butt joint.
8	Plumbing: Demonstration and Practice of Plumbing tools, Preparation of Pipe joints with coupling for same diameter and with reducer for different diameters.
9	Assembly and dis assembly of Two-wheeler vehicle
10	Tinkering Lab experiments: a) Build and demonstrate automatic streetlight using LDR. b) Simulate the AURDINO LED blinking activity in Tinker cad c) Interface a basic sensor and actuator with a microcontroller
Reference Books:	
1	Elements of Workshop Technology, Vol. I by S. K. Hajra Choudhury & Others, Media Promoters and Publishers, Mumbai. 2007, 14th edition
2	Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004.
3	Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P.A.; Atul Prakashan, 2021-22.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610302	BS	--	--	3	1.5	50	50	3 Hrs.

PHYSICS FOR MECHANICAL SYSTEMS LAB

(For ME)

Course Objectives:

1	To impart hands on experience to the students entering Engineering & Technology education about handling equipment/instruments and use them in experimentation.
2	To make the students understand theoretical aspects of various phenomena experimentally.

Course Outcomes:

S.No.	Outcome	Knowledge Level
1	Apply fundamental principles of Physics to conduct engineering laboratory experiments.	K3
2	Get hands on experience in setting up experiments and using the instruments/equipment individually and conduct experiments.	K3
3	Apply graphical and analytical methods to analyze experimental data and determine physical parameters.	K3
4	Get introduced to using new/advanced technologies and understand their significance .	K3

List of Experiments

(Any **10** of the following listed experiments)

(Out of which any **TWO** experiments may be conducted in virtual mode)

1	Determination of thickness of thin object by wedge method.
2	Determination of radius of curvature of a given plano convex lens by forming Newton's rings.
3	Determination of wavelengths of different spectral lines in mercury spectrum using diffraction grating in normal incidence configuration.
4	Determination of resolving power of telescope.
5	Determination of dielectric constant of given capacitor by charging and discharging method.
6	Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
7	Determination of temperature coefficients of a thermistor.
8	Determination of wavelength of Laser and grating element.
9	Estimation of Planck's constant using photoelectric effect.
10	Determination of the Resistivity of semiconductor by four probe method.
11	Determination of the energy gap of a semiconductor using p-n junction diode.
12	Study the variation of magnetic field along the axis of a current carrying circular coil by Stewart & Gee's Method
13	Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall Effect.
14	Determination of Acceleration due to gravity and Radius of gyration using Compound pendulum.

15	Determination of magnetic susceptibility by Kundt's tube method
16	Verification of Brewster's law.
17	Determination of frequency of A.C. supply using Sonometer.
18	Determination of frequency of electrically maintained tuning fork by Melde's apparatus.
19	Determination of Rigidity modulus of material of a wire-dynamic method (Torsional pendulum).
Reference Books:	
1	S. Balasubramanian, M.N. Srinivasan "A Text Book of Practical Physics"- S Chand Publishers, 2017.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610303	**HS**	**--**	**--**	**3**	**1.5**	**50**	**50**	**3 Hrs**

English for Technical Communication and MOODLE Laboratory

(For ME)

Course Objectives:

1	Enable students apply correct phonological patterns for effective communication.
2	Develop students' ability to employ listening strategies to comprehend audio texts.
3	Enable students participate effectively in group discussions, presentations, debates, and interviews in professional contexts.
4	Enhance students' ability to write cohesively and coherently.
5	Familiarize students with Moodle LMS and digital learning tools for effective use of learning resources

Course Outcomes:

S. No	Outcome	Knowledge Level
1	Apply appropriate phonological features to communicate effectively in formal and informal contexts.	K3
2	Apply listening strategies to comprehend audio texts, take notes, and narrate events.	K3
3	Demonstrate effective participation in group discussions, presentations, debates, and interviews to communicate in professional contexts.	K3
4	Develop an ability to write clearly on topics related to technology and workplace communication.	K3
5	Demonstrate proficiency in using Moodle LMS and digital learning tools to access learning resources effectively.	K2

SYLLABUS

PART-A

1	Phonetics & Accent Neutralization
2	Introducing Self and Others
3	Brief Talk in Formal/Informal Contexts, JAM/Extempore
4	Listening Comprehension
5	Audio-based Note Taking & Narrating Events
6	Process Description and Paragraph Writing (Technical Lab Context)
7	Group Discussions and Deliberations
8	Micro Presentations (2-5 Mts)
9	Presenting Logical Arguments – Debates
10	Interviews – Formal & Semi-formal

PART-B Moodle Learning Management System: Digital Learning and Academic Skills	
1	Introduction to Moodle LMS • Introduction to Moodle • Student login and profile management • Moodle dashboard and course navigation • Accessing announcements and course materials • Notifications and calendars
2	Learning Through Moodle • Accessing PDFs, videos, web resources, and lecture materials • Downloading and organizing study materials • Moodle lessons and learning activities • Tracking course completion • Managing deadlines and study schedules
3	Assignments, Quizzes and Assessments • Understanding assignment instructions • Creating and uploading assignments • File formats and file naming • Online quizzes and question types • Viewing marks and teacher feedback • Resubmission and revision
4	Digital Communication and Collaboration • Moodle forums and discussions • Academic email etiquette • Online communication and netiquette • Group activities and peer learning • Collaborative learning tools • Digital presentation skills
Lab Manual:	
1	Propel: A Course Book for Technical English Communication. Cengage Publishers, 2026.
Reference Books:	
1	Richards, Jack C., Jonathan Hull, and Susan Proctor. Interchange Level 3 Student's Book with Self-study DVD-ROM. Vol. 3. Cambridge University Press, 2012.
2	Taylor Grant: English Conversation Practice,Tata McGraw Hill Education India,2016.
3	Hewing's, Martin. Cambridge Academic English (B2).CUP, 2012.
4	J.Sethi & P.V.Dhamija. A Course in Phonetics and Spoken English, (2ndEd), Kindle, 2013
5	Raman Meenakshi,Sangeeta-Sharma.TechnicalCommunication.OxfordPress.2018.

Web Resources:	
	1. www.esl-lab.com 2. www.englishmedialab.com 3. www.englishinteractive.net 4. https://www.britishcouncil.in/english/online 5. http://www.letstalkpodcast.com 6. https://www.youtube.com/c/mmmEnglish_Emma/featured 7. https://www.youtube.com/c/ArnelsEverydayEnglish/featured 8. https://www.youtube.com/c/engvidAdam/featured 9. https://www.youtube.com/c/EnglishClass101/featured 10. https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists 11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw 12. https://www.youtube.com/user/letstalkaccent/videos 13. https://www.youtube.com/c/EngLanguageClub/featured 14. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc 15. https://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610017	**MC**	**2**	**--**	**--**	**--**	**50**	**--**	**--**

Universal Human Values – II

(Common to AIML, CSE, CSIT, IT, MECH)

Course Objectives: The objective of this course is to make the student aware of

1.	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2.	Harmony in the human being, family, society and nature/existence
3.	Holistic perspective towards life, profession and happiness

Course Outcomes: At the end of this course student will be able to

S.No	Outcome	Knowledge Level
1.	**Explain** the role of value education in achieving basic human aspirations.	K2
2.	**Summarize** needs to obtain harmony in self (I).	K2
3.	**Describe** criteria for human-human relationship and harmony in society.	K2
4.	**Explain** four orders of nature and our existence.	K2
5.	**Interpret** significance of harmony in holistic development	K2

SYLLABUS

UNIT-I (06Hrs)	**Introduction to Value Education**: Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education); Understanding Value Education-Practice Session PS1 Sharing about Oneself; self-exploration as the Process for Value Education; Continuous Happiness and Prosperity – the Basic Human Aspirations-Practice Session PS2 Exploring Human Consciousness; Happiness and Prosperity – Current Scenario; Method to Fulfill the Basic Human Aspirations-Practice Session PS3 Exploring Natural Acceptance.
UNIT-II (08Hrs)	**Harmony in the Human Being**: Understanding Human being as the Co-existence of the self and the body; Distinguishing between the Needs of the self and the body -Practice Session PS4 Exploring the difference of Needs of self and body; The body as an Instrument of the self; Understanding Harmony in the self-Practice Session PS5 Exploring Sources of Imagination in the self; Harmony of the self with the body;

	Programme to ensure self-regulation and Health - Practice Session PS6 Exploring Harmony of self with the body.
UNIT-III (06 Hrs)	**Harmony in the Family and Society**: Harmony in the Family – the Basic Unit of Human Interaction; 'Trust' – the Foundational Value in Relationship-Practice Session PS7 Exploring the Feeling of Trust; 'Respect' – as the Right Evaluation-Practice Session PS8 Exploring the Feeling of Respect; Other Feelings, Justice in Human-to-Human Relationship; Understanding Harmony in the Society; Vision for the Universal Human Order- Practice Session PS9 Exploring Systems to fulfil Human Goal.
UNIT-IV (06 Hrs)	**Harmony in the Nature/Existence:** Understanding Harmony in the Nature; Interconnectedness, self- regulation and Mutual Fulfilment among the Four Orders of Nature-Practice Session PS10 Exploring the Four Orders of Nature; Realizing Existence as Co-existence at All Levels; The Holistic Perception of Harmony in Existence-Practice Session PS11 Exploring Co-existence in Existence.
UNIT-V (06 Hrs)	**Implications of the Holistic Understanding – a Look at Professional Ethics**: Natural Acceptance of Human Values; Definitiveness of (Ethical) Human Conduct- Practice Session PS12 Exploring Ethical Human Conduct; A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order; Competence in Professional Ethics-Practice Session PS13 Exploring Humanistic Models in Education; Holistic Technologies, Production Systems and Management Models-Typical Case Studies; Strategies for Transition towards Value-based Life and Profession-Practice Session PS14 Exploring Steps of Transition towards Universal Human Order.
Textbooks:	
1.	The Textbook R R Gaur, R Asthana, G P Bagaria, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1
2.	The Teacher's Manual R R Gaur, R Asthana, G P Bagaria,*Teachers' Manual for A Foundation Course in Human Values and Professional Ethics,* 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2

Reference Books:	
1.	*JeevanVidya: EkParichaya*, A Nagaraj, JeevanVidyaPrakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth – by Mohandas Karamchand Gandhi
5.	Small is Beautiful – E. F Schumacher.
6.	Slow is Beautiful – Cecile Andrews
7.	Economy of Permanence – J C Kumarappa
8.	Bharat Mein Angreji Raj – PanditSunderlal
9.	Rediscovering India – by Dharampal
10.	*Hind Swaraj or Indian Home Rule* – by Mohandas K. Gandhi
11.	*India Wins Freedom* – Maulana Abdul Kalam Azad
12.	*Vivekananda* – Romain Rolland (English)
13.	*Gandhi* – Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%201-Introduction%20to%20Value%20Education.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%202-Harmony%20in%20the%20Human%20Being.pdf
3.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%203-Harmony%20in%20the%20Family.pdf
4.	https://fdp-si.aicte-india.org/UHV%201%20Teaching%20Material/D3-S2%20Respect%20July%2023.pdf
5.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%205-Harmony%20in%20the%20Nature%20and%20Existence.pdf
6.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf
7.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
8.	https://www.studocu.com/in/document/kiet-group-of-institutions/universal-human-values/chapter-5-holistic-understanding-of-harmony-on-professional-ethics/62490385
9.	https://onlinecourses.swayam2.ac.in/aic22_ge23/preview

Regulation:R26	I/IV-B.Tech. II -Semester

MECHANICAL ENGINEERING

COURSE STRUCTURE

(With effect from 2026-27 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
A2620001	Differential Equations and Vector Calculus	**BS**	3	0	0	3	40	60	100
A2620301	Chemistry for Mechanical Engineers	**BS**	3	0	0	3	40	60	100
A2620007	Computational Thinking and Problem Solving using Python	**ES**	3	0	0	3	40	60	100
A2620302	Manufacturing Processes	**PC**	3	0	0	3	40	60	100
A2620009	Engineering Graphics	**ES**	0	1	3	2.5	50	50	100
A2620016	Computational Thinking and Problem Solving using Python lab	**ES**	0	0	3	1.5	50	50	100
A2620303	Manufacturing Processes lab	**PC**	0	0	3	1.5	50	50	100
A2620304	Chemistry for Mechanical Engineers lab	**BS**	0	0	3	1.5	50	50	100
A2620305	Basic Electrical and Electronics Engineering Lab	**ES**	0	0	3	1.5	50	50	100
A2620021	NSS, Sports & Yoga	**MC**	0	0	2	0	50	-	50
		TOTAL	**12**	**1**	**17**	**20.5**	**460**	**490**	**950**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620001	BS	3	--	--	3	40	60	3 Hrs.

DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

(Common for all Branches)

Course Objectives: Students are expected to learn

1.	First order ordinary differential equations and some simple geometrical and physical applications.
2.	Methods of solution of linear higher order ordinary differential equations.
3.	Formation and solution of linear partial differential equations.
4.	Concepts of Gradient, divergence, curl.
5.	Vector integral theorems.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.	K3
2.	**Solve** linear ordinary differential equations of second and higher order which occur in various engineering fields.	K3
3.	**Apply** the appropriate methods of solution for partial differential equations that model physical processes.	K3
4.	**Compute** the gradient of a scalar field, curl and divergence of various vector fields to demonstrate their physical behavior.	K3
5.	**Calculate** the work done against a field, circulation and flux using vector calculus.	K3

SYLLABUS

UNIT-I (10Hrs)	**Differential equations of first order and first degree** Exact equations and equations reducible to exact form (Integrating Factor), Linear differential equations, Bernoulli's equations. Orthogonal Trajectories, Applications: Newton's Law of cooling, Law of natural growth and decay, Electrical circuits: RL & RC circuits.
UNIT-II (10 Hrs)	**Linear differential equations of higher order (Constant Coefficients)** Definitions, homogenous and non-homogenous, complimentary function, particular integral, general solution, Method of variation of parameters – Wronskian. Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion.
UNIT-III (10 Hrs)	**Partial Differential Equations** Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous and Non-Homogeneous Linear Partial differential equations with

	constant coefficients.
UNIT-IV (10 Hrs)	**Vector Differentiation** Scalar and vector point functions, vector operator Del, Gradient and applications, Directional derivative, Divergence – Solenoidal and Curl – Irrotational Field – Scalar Potential, Vectoridentities.
UNIT-V (10 Hrs)	**Vector Integration** Line Integral – Circulation – Work done by force, surface integral – flux, Green's theorem in the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and their applications.

Textbooks:	
1.	HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition.
2.	AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition.
Reference Books:	
1.	Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2.	Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3.	Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4.	Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5.	Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017
6.	Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty two Edition (Reprint 2022).
7.	Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015.
e-Resources	
1.	https://mathworld.wolfram.com/First-OrderOrdinaryDifferentialEquation.html
2.	https://en.wikipedia.org/wiki/Differential_equation
3.	https://en.wikipedia.org/wiki/Partial_differential_equation
4.	https://en.wikipedia.org/wiki/Vector_calculus
5.	https://en.wikipedia.org/wiki/Vector_calculus

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620301	BS	3	--	--	3	40	60	100

CHEMISTRY FOR MECHANICAL ENGINEERS

(For ME)

Course Objectives:

1.	Understand corrosion mechanisms and select suitable corrosion-control and surface-engineering methods for mechanical components.
2.	Examine the properties and selection criteria of lubricants and refractories used in machines and high-temperature systems
3.	Apply fuel and energy-chemistry principles to evaluate conventional and green fuels for engineering applications
4.	Relate the composition, processing, properties, and applications of engineering materials, polymers, ceramics, and composites to mechanical-design needs.
5.	Assess the potential of advanced materials, including nanomaterials, high-entropy alloys, smart materials, and self-healing materials, for modern mechanical engineering applications.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	**Apply** electrochemical corrosion principles to identify corrosion forms and recommend appropriate cathodic/anodic protection, coatings, and surface-treatment methods for mechanical components.	K3
2.	**Apply** lubricant-property data and refractory characteristics to select materials for friction control, wear reduction, and high-temperature service.	K3
3.	**Analyze** fuel characteristics, calorific values, and fuel-cell options to select suitable conventional or green fuels for engineering applications.	K3, K4
4.	**Analyze** the composition, processing, heat treatment, properties, and applications of metallic alloys, polymers, ceramics, elastomers, and composites for mechanical systems.	K3, K4
5.	**Apply the knowledge** the suitability of nanomaterials, graphene-based composites, high-entropy alloys, smart materials, self-healing materials, and magnetic materials for advanced mechanical-engineering applications.	K3, K4

SYLLABUS

UNIT-I (10Hrs)	**Corrosion Chemistry and Protective Methods** • Chemical and electrochemical theories of corrosion, pilling Bed worth's rule, galvanic series, Types of corrosion (galvanic, pitting, stress corrosion), factors affecting the rate of corrosion. Corrosion prevention: cathodic protection (sacrificial anodic and impressed-current protection). • Protective coatings: Galvanizing (hot-dip galvanizing, electro-galvanizing).

	• Electroplating (copper and nickel plating), Electroless plating, Anodizing: aluminium anodizing, surface hardening: carburizing, nitriding, carbonitriding, conversion coatings: phosphating and chromating. Corrosion penetration rate (CPR)
UNIT-II (10 Hrs)	**Chemistry of Lubricants and Refractories** • Lubricants: Functions of lubricants - mechanism [fluid-film (hydrodynamic) lubrication, boundary-film (thin-film) lubrication, extreme-pressure lubrication], classification (solid, semisolid, liquid), Properties of lubricants — viscosity, viscosity index, flashpoint, fire point, cloud point, pour point, aniline point, saponification. • Refractories: Characteristics, classification, properties (refractoriness, refractories under load, thermal spalling, porosity) - factors affecting the refractory materials - applications
UNIT-III (10 Hrs)	**Energy Chemistry** • Classification, calorific value, HCV and LCV, Dulong's formula, problems; calorific-value determination using a bomb calorimeter, problems. • Solid fuels (coal): Classification (anthracite, bituminous, lignite, peat), analysis (proximate, ultimate). • Liquid fuels (petroleum oil): Composition, refining processes, fractional distillation, synthetic fuels (Fischer-Tropsch synthesis). • Green fuels: Synthesis of Green hydrogen, hythane. • Knocking in petrol engines: Octane number, anti-knock additives (TEL, MTBE, ethanol); knocking in diesel engines: cetane number, ignition quality, anti-knock additives. • Fuel cell: Introduction, H_2-O_2 fuel cells, applications
UNIT-IV (10 Hrs)	**Engineering Materials and Composites** • Ferrous alloys: Composition and applications of carbon steels, alloy steels, stainless steels, cast irons; heat treatment — annealing, normalizing, hardening, tempering, age hardening. • Polymer chemistry: Types of polymerization, differences between thermoplastics, thermosets; engineering properties and applications of ABS, Bakelite; polymer processing (injection moulding, extrusion moulding, compression moulding). • Elastomers: Natural rubber, vulcanization, synthetic rubbers – (Preparations, properties and applications of BUNA-S and BUNA-N). • Ceramics: Properties and applications of Alumina, silicon carbide, silicon nitride, zirconia. • Composite materials: Classification and applications
UNIT-V (10 Hrs)	**Advanced Materials & Nanochemistry** • Introduction to nanomaterials, carbon nanotubes (CNTs)-synthesis by chemical

	vapor deposition (CVD) method, sol-gel process (metal oxides nanoparticles), applications. • Nanomaterials for Mechanical Engineering: Graphene, Graphene-Based Composites (types, properties)- Applications (Wear and Friction Reduction). • High-Entropy Alloys (HEAs), Smart Materials and Shape-Memory Alloys: Definition, properties, applications. • Self-Healing Materials: Types, working, applications.
Textbooks:	
1.	Jain, P. C., & Jain, M. (2020). Engineering chemistry (17th ed.). Dhanpat Rai Publishing Company.
2.	Agarwal, S. (2026). Engineering chemistry (5th ed.). Cambridge University Press.
Reference Books:	
1.	Davis, M. L. (2020). Water and wastewater engineering: Design principles and practice (2nd ed.). McGraw-Hill Education.
2.	Shikha, A., & Sharma, R. K. (2019). Engineering chemistry. Krishna Prakashan Media.
3.	Dara, S. S., &Umare, S. S. (2018). Engineering chemistry (15th ed.). S. Chand Publishing.
e-Resources	
1.	Society of Tribologists and Lubrication Engineers. (n.d.). Tribology and lubrication resources. https://www.stle.org
2.	ASM International. (n.d.). Materials information and education. https://www.asminternational.org
3	Society of Automotive Engineers. (n.d.). Automotive engineering standards. https://www.sae.org
4	American Society of Mechanical Engineers. (n.d.). Mechanical engineering resources. https://www.asme.org
5	National Association of Corrosion Engineers. (n.d.). Corrosion prevention and control. https://www.nace.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620007	ES	3	--	--	3	40	60	3 Hrs.

Computational Thinking and Problem Solving Using Python

(Common for CE & ME)

Course Objectives:

1	Establish computational thinking, decomposition, pattern recognition, abstraction, and algorithmic reasoning as the foundation for problem-solving, with strong grounding in algorithm design and flowchart construction.
2	Develop proficiency in Python control structures while explicitly linking conditionals to abstraction and loops to pattern recognition.
3	Build competence in selecting Python data structures (strings, lists, tuples, dictionaries) as an exercise in abstraction and pattern-based organization of information.
4	Train students to decompose problems into functions and recursive sub-problems, reason informally about algorithmic efficiency, and use AI coding assistants responsibly.
5	Equip students with file handling, exception handling, and basic data analysis/visualization skills framed as decomposing a larger data workflow into manageable, robust steps and to integrate these skills in a capstone mini-project.

Course Outcomes: At the end of the course, student will be able to

S.No	Out comes	Knowledge Level
1	Explain the principles of computational thinking and interpret algorithms and flowcharts that represent problem-solving logic	K2
2	Apply pattern recognition and abstraction, via Python control structures, to implement step-by-step algorithmic solutions	K3
3	Analyze problems to select and manipulate appropriate data structures (strings, lists, tuples, dictionaries) as an abstraction and organization exercise	K4
4	Examine the correctness and efficiency of decomposed, modular algorithms and Python programs through tracing, testing, debugging, and exception handling	K4
5	Analyze modular, well-structured Python programs by decomposing problems into functions, recursion, file handling, and data analysis, culminating in an integrative capstone project	K4

UNIT-I (10 Hrs)	Computational Thinking, Algorithm Design & Flowcharting Computational thinking pillars - decomposition, pattern recognition, abstraction, algorithmic thinking, and the problem-solving process. Algorithm design - characteristics of a good algorithm, stepwise refinement, pseudocode. Flowcharts - standard symbols, construction rules, and manual tracing for verification.

	Translating algorithms and flowcharts into Python code - variables, data types, operators, and input/output. Introduction to good coding practice - docstrings, inline comments, and print-based debugging, threaded through every unit from here on
UNIT – 2 **(10 Hrs)**	Control Structures, Conditional statements - as abstraction over multiple cases. Loops (for, while) - as pattern recognition of repetition. Nested loops and loop control (break, continue, pass) - as decomposition of complex repetitive logic.
UNIT – 3 **(10 Hrs)**	Data Structures — Strings, Lists, Tuples &Dictionaries Strings - pattern matching and abstraction of text as a sequence.Lists - abstraction of ordered, mutable collections.Tuples and dictionaries - abstraction of fixed and key-based data
UNIT – 4 **(10 Hrs)**	Functions & Modular ProgrammingFunctions - decomposition of a program into independent, reusable sub-problems. Recursion - algorithmic thinking applied to self-similar sub-problems.Basic algorithmic efficiency - informal comparison of iterative vs. recursive approaches and their time trade-offs. Modules and built-in libraries, including responsible AI-assisted coding practices.
UNIT – 5 **(10 Hrs)**	File Handling, Exception Handling, Data Analysis & Capstone Project File handling - decomposition of a data workflow into read/process/write stages. Exception handling - abstraction of error cases into predictable, manageable patterns. Introduction to data analysis and visualization (NumPy/Pandas/Matplotlib basics) - pattern recognition in real datasets. Capstone integration - combining all prior units into one end-to-end project.
Text Books:	
1	Downey, A. B. (2015). Think Python: How to think like a computer scientist (2nd ed.). O'Reilly Media
2	Guttag, J. V. (2021). Introduction to computation and programming using Python: With application to computational modeling and understanding data (3rd ed.). MIT Press.
Reference Books:	
1	Sweigart, A. (2025). Automate the boring stuff with Python: Practical programming for total beginners (3rd ed.). No Starch Press
2	McKinney, W. (2022). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media.
3	Wing, J. M. (2006). Computational thinking. Communications of the ACM, 49(3), 33–35. https://doi.org/10.1145/1118178.1118215
Web References:	
1	Python Software Foundation. (n.d.). The Python tutorial. https://docs.python.org/3/tutorial/
2	W3Schools. (n.d.). Python tutorial. https://www.w3schools.com/python/

3	NPTEL. (n.d.). Programming, data structures and algorithms using Python. https://nptel.ac.in

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620302	PC	3	--	--	3	40	60	3 Hrs.

MANUFACTURING PROCESSES

(For ME)

Course Objectives:

1.	Know the working principle of different metal casting processes and gating system.
2.	Classify the welding processes, working of different types of welding processes and welding defects.
3.	Know the nature of plastic deformation, cold and hot working process, working of a rolling mill and types, extrusion processes.
4.	Understand the principles of forging, tools and dies, working of forging processes.
5.	Know about the Additive manufacturing and coating technologies.

Course Outcomes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** the principles of metal casting for manufacturing various mechanical components.	K3
2.	**Classify** Various types of welding processes and welding defects.	K3
3.	**Apply** the Knowledge of metal forming to cold and hot worked components.	K3
4.	**Identify** suitable tools and dies for various forging processes.	K3
5.	**Identify** suitable Additive manufacturing process for manufacturing various mechanical components and different coating technologies.	K3

SYLLABUS

UNIT-I (10Hrs)	**Casting:** Steps involved in making a casting – Advantage of casting. Types of patterns – Materials used for patterns, pattern allowances. Cupola furnace, Casting defects- causes and remedies. Basic principles and applications of special casting processes - Centrifugal casting, Die casting, Investment casting, Permanent mould casting and shell molding.
UNIT-II (10 Hrs)	**Welding:** Classification of welding processes, types of welded joints and their characteristics, Gas welding, Different types of flames, Basic principles of Arc welding, submerged arc welding, TIG & MIG welding, Resistance welding, Friction welding, Forge welding, Plasma Arc welding, Laser welding, electron beam welding, Soldering & Brazing. Welding defects – causes and remedies.
UNIT-III (10 Hrs)	**Bulk Forming:** Plastic deformation in metals and alloys-recovery, recrystallization. Hot working and Cold Working. Bulk forming processes: Forging-Types of Forging, forging defects and remedies; Rolling – fundamentals, types of rolling mills. Extrusion -Types of extrusion, Impact extrusion, Hydrostatic extrusion; Wire drawing and

	Tube drawing.
UNIT-IV (10 Hrs)	**Sheet metal forming:** Blanking and piercing, Deep drawing, Stretch forming, Bending, Spring back and its remedies, Coining, Spinning, Types of presses and press tools. High energy rate forming processes: Principles of explosive forming, electromagnetic forming, Rubber pad forming, advantages and limitations.
UNIT-V (10 Hrs)	**Additive manufacturing:** Steps in Additive Manufacturing (AM), Classification of AM processes, Advantages of AM, VAT photo polymerization AM Processes, Extrusion - Based AM Processes, Applications. **Coating Technologies:** Thermal spray coatings-- plasma spraying and arc spraying. Vapour-phase coatings-- physical vapour deposition(PVD) and chemical vapour deposition(CVD).

Textbooks:	
1.	Kalpak Jain S and Steven R Schmid, Manufacturing Processes for Engineering Materials, 5/e, Pearson Publications, 2007.
2.	P.N. Rao, Manufacturing Technology -Vol I, 5/e, McGraw Hill Education, 2018.
3.	J.R. Davis, Surface Engineering For Corrosion and Wear resistance, ASM International, 2001
Reference Books:	
1.	A. Ghosh & A.K. Malik, Manufacturing Science, East West Press Pvt. Ltd, 2010.
2.	Lindberg and Roy, Processes and materials of manufacture, 4/e, Prentice Hall India Learning Private Limited, 1990.
3.	R.K. Jain, Production Technology, Khanna Publishers, 2022.
4.	Sharma P.C., A Text book of Production Technology, 8/e, S Chand Publishing, 2014.\
5.	H.S. Shaun, Manufacturing Processes, 1/e, Pearson Publishers, 2012.
6.	WAJ Chapman , Workshop Technology, 5/e, CBS Publishers & Distributors Pvt. Ltd, 2001.
7.	Hindustan Machine Tools, Production Technology, Tata McGraw Hill Publishers, 2017.
8.	Ian Gibson, David W Rosen, Brent Stucker., Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing, 2/e, Springer, 2015.
e-Resources	
1.	https://www.edx.org/learn/manufacturing/massachusetts-institute-of-technology-fundamentals-of-manufacturing-processes
2.	https://onlinecourses.nptel.ac.in/noc21_me81/preview
3.	https://www.coursera.org/learn/introduction-to-additive-manufacturing-processessera
4.	https://archive.nptel.ac.in/courses/112/103/112103263/
5.	https://elearn.nptel.ac.in/shop/nptel/principles-of-metal-forming-technology/?v=c86ee0d9d7ed

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620009	ES	--	1	3	2.5	50	50	3 Hrs

ENGINEERING GRAPHICS

(Common to CE, EEE & ME)

Course Objectives:

1	Develop spatial visualization and graphical communication skills to represent three-dimensional engineering objects using standard engineering graphics principles.
2	Create 2D, 3D CAD models and communicate engineering design information effectively through conventional drawings and digital CAD tools.

Course Outcomes:

S.No	Outcome	Knowledge Level
1	**Apply** standard engineering graphics principles and conventions to prepare accurate sketches and engineering drawings.	**K3**
2	**Visualize** and represent three-dimensional objects and systems using appropriate projection, orthographic, isometric, sectional, and surface-development techniques.	**K4**
3	Use CAD tools to **Create**, modify, and **Communicate** 2D drawings and 3D models with appropriate dimensions, annotations, symbols, and features.	**K4**
4	**Generate** complete engineering design documentation through original graphical representations, detailed CAD models, and compliant engineering drawings for multidisciplinary applications.	**K4**

SYLLABUS

Section A – Manual Engineering Drawing

1. **Geometrical Constructions and Engineering Curves:**
   Introduction to Engineering Drawing, Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods.
2. **Engineering Curves**: Parabola, Ellipse and Hyperbola by general method (Eccentricity method only), Cycloidal curves, Involutes, tangent & normal for these curves.
3. **Projection of Points and Straight Lines:** Projection of points in different quadrants and straight lines in different orientations.
4. **Projections of planes:** projections of regular planes perpendicular to one reference plane and parallel or inclined to the other, Planes inclined to both reference planes.
5. **Projections of Solids**: Types of solids- Polyhedra and Solids of revolution. Projections of solids in simple positions.
6. **Sections of Solids**: Sections and Sectional views of Right and Regular Solids.
7. **Isometric Projection:** Introduction to Isometric projection and Isometric projection of simple Right and Regular Solids – Prism, Cylinder, Pyramid and Cone.

Section B –Computer Aided Engineering Graphics

1. **Computer-Aided Engineering Graphics (CAD)** Introduction to CAD software, standard tool bar menu and description of most commonly used tool bars, and navigational tools.
2. **CAD Geometrical Constructions and Curves**: Construction of polygons, circles, arcs, ellipse and other engineering curves using CAD.
3. **2D CAD – Orthographic Views**: Preparation of front, top, side views of simple engineering objects
4. **Basic 3D CAD Modelling:** Creation of simple three-dimensional engineering objects using basic modelling features.
5. **3D CAD Modelling with Features:** Creation and modification of 3D models using features such as hole, fillet, chamfer, mirror, pattern and other appropriate tools
6. **Construct a 3D CAD model** of a given engineering object from its front, top and side orthographic views.
7. **CAD Sectional Views:** Creation of sectional views of 3D engineering objects and generation of corresponding 2D drawings.
8. **CAD Surface Development**: Development of cylindrical, conical, prismatic surfaces using CAD.
9. **3D Model to Engineering Drawing:** Generation of orthographic, isometric and sectional views with dimensions, annotations and symbols from a 3D CAD model.
10. **CAD Modelling of Physical Engineering Objects:** Measurement and interpretation of a physical engineering object, recreation of its geometry in CAD, and preparation of its engineering drawing.

Reference Books:

1	Bhatt, N. D., & Panchal, V. M. (2023). Elementary engineering drawing (53rd ed.). Charotar Publishing House.
2	Narayana, K. L., Kannaiah, P., & Reddy, K. V. (2020). Engineering graphics (2nd ed.). Scitech Publications
3	Venugopal, K., & Prabhu Raja, V. (2019). Engineering drawing and graphics (5th ed.). New Age International.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620016	ES	--	--	3	1.5	50	50	3 Hrs.

Computational Thinking and Problem Solving Using Python Lab

(Common to CE & ME)

Course Objectives: At the end of the course, the teacher should be able to

1	Establish computational thinking, decomposition, pattern recognition, abstraction, and algorithmic reasoning as the foundation for problem-solving, with strong grounding in algorithm design and flowchart construction.
2	Develop proficiency in Python control structures while explicitly linking conditionals to abstraction and loops to pattern recognition.
3	Build competence in selecting Python data structures (strings, lists, tuples, dictionaries) as an exercise in abstraction and pattern-based organization of information.
4	Train students to decompose problems into functions and recursive sub-problems, reason informally about algorithmic efficiency, and use AI coding assistants responsibly.
5	Equip students with file handling, exception handling, and basic data analysis/visualization skills framed as decomposing a larger data workflow into manageable, robust steps and to integrate these skills in a capstone mini-project.

Course Outcomes: At the end of the course, student will be able to

S.No	Out comes	Knowledge Level
1	Explain the principles of computational thinking and interpret algorithms and flowcharts that represent problem-solving logic	K2
2	Apply pattern recognition and abstraction, via Python control structures, to implement step-by-step algorithmic solutions	K3
3	Analyze problems to select and manipulate appropriate data structures (strings, lists, tuples, dictionaries) as an abstraction and organization exercise	K4
4	Evaluate the correctness and efficiency of decomposed, modular algorithms and Python programs through tracing, testing, debugging, and exception handling	K5
5	Create modular, well-structured Python programs by decomposing problems into functions, recursion, file handling, and data analysis, culminating in an integrative capstone project	K6

1	**Week 1 (Unit 1):** 1. Find the largest of three numbers. 2. Check whether a given year is a leap year
2	**Week 2 (Unit 1):** (1) Compute simple interest given principal, rate, and time. (2) Compute the area and perimeter of a rectangle from user input.

	(3) Swap two numbers without using a third variable.
3	**Week 3 (Unit 2):** (1) Determine whether a number is positive, negative, or zero. (2) Check if a given number is prime (3) Convert temperature between Celsius and Fahrenheit based on user choice (4) Determine BMI category (underweight/normal/overweight/obese) using elif. (5) Classify a triangle as equilateral, isosceles, or scalene.
4	**Week 4 (Unit 2):** (1) Print all Armstrong numbers between 1 and 1000. (2) Generate the Fibonacci series up to n terms using a loop (3) Find the GCD and LCM of two numbers using a while loop (4) Simulate a simple ATM PIN-entry system allowing a maximum of 3 attempts. Find the first n prime numbers using nested loops with break
5	**Week 5 (Unit 3):** (1) Encrypt a message using a Caesar cipher. (2) Check if two given strings are anagrams of each other. (3) Check whether a given string is a palindrome using slicing and string methods (4) Count vowels, consonants, digits, and spaces in a given string
6	**Week 6 (Unit 3):** (1) Find the second-largest element in a list of exam scores. (2) Merge two sorted lists of student roll numbers into a single sorted list. (3) Generate squares of all even numbers from 1 to 50 using list comprehension. (4) Store and retrieve student records using a dictionary (roll number as key, tuple of name/marks as value
7	**Week 7 (Unit 3):** (1) Store coordinates (x, y) as tuples in a list and compute the distance between consecutive points. (2) Count the frequency of each word in a paragraph using a dictionary (3) Maintain an inventory system (item name as key, tuple of price and quantity as value) that updates stock after each transaction.
8	**Week 8 (Unit 4):** (1) Using a function, check if a number is "perfect," and list all perfect numbers up to 10,000. (2) Using *args, compute the total resistance of resistors in series or parallel. (3) Define a function with default/keyword arguments to compute the area of different shapes based on shape type and use it. (4) Write a function that returns the mean, median, and mode of a list of number
9	**Week 9 (Unit 4):** (1) Compute the GCD of two numbers recursively using the Euclidean algorithm. (2) Compute factorial using recursion (3) Solve the Tower of Hanoi problem for n disks recursively and count the number of moves. (4) Compute the range and maximum height of a projectile given initial velocity and angle, using the math module
10	**Week 10 (Unit 5):** (1) Count the words, lines, and characters in a given text file.

	(2) Read a log file of sensor readings and append a new set of readings with a timestamp. (3) Read a text file of daily temperature readings and write the maximum, minimum, and average to an output file
11	**Week 11 (Unit 5):** (1) Build a simple calculator that gracefully handles ZeroDivisionError, ValueError, and invalid operators. (2) Safely compute the roots of a quadratic equation using try-except-finally, abstracting invalid input and division-by-zero into predictable, handled error patterns. (3) Read a CSV of student marks and write the names of students scoring above 75% to a new file (4) Validate user input for a unit-conversion tool and raise a custom exception for out-of-range values. (5) Using Pandas, read a CSV of daily electricity consumption and plot a line graph over a month, recognizing weekday/weekend consumption patterns
12	**Week 12 (Unit 5):** (1) Capstone Mini-Project: "Dataset Health-Check & Insights Tool" — decompose the workflow into read → validate → clean → analyze → visualize → report stages, using exception handling and NumPy/Pandas/Matplotlib, submitted with a one-page reflection.
Text Books:	
1	Downey, A. B. (2015). Think Python: How to think like a computer scientist (2nd ed.). O'Reilly Media.
2	Guttag, J. V. (2021). Introduction to computation and programming using Python: With application to computational modeling and understanding data (3rd ed.). MIT Press
Reference Books:	
1	Sweigart, A. (2025). Automate the boring stuff with Python: Practical programming for total beginners (3rd ed.). No Starch Press.
2	McKinney, W. (2022). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media.
3	Wing, J. M. (2006). Computational thinking. Communications of the ACM, 49(3), 33–35. https://doi.org/10.1145/1118178.1118215
Web References:	
1	Python Software Foundation. (n.d.). The Python tutorial. https://docs.python.org/3/tutorial/
2	W3Schools. (n.d.). Python tutorial. https://www.w3schools.com/python/
3	NPTEL. (n.d.). Programming, data structures and algorithms using Python. https://nptel.ac.in

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620303	PC	--	--	3	1.5	50	50	3 Hrs.

MANUFACTURING PROCESSES LAB

(For ME)

Course Objectives:

1.	To impart hands-on practical exposure on manufacturing processes and equipment.

Course Outcomes: At the end of the course students will be able to

S. No	Outcome	Knowledge Level
1.	**Apply** the knowledge of casting to make various sand moulds.	K3
2.	**Determine** various moulding sand properties.	K3
3.	**Apply** a suitable welding process for preparing a component.	K3
4.	**Apply** the knowledge of forming & forging to prepare various components.	K3
5.	**Demonstrate** the procedures of 3D printing, metal Castings, sintering and CVD	K3

SYLLABUS

1.	Prepare a sand mould of stepped pulley.
2.	Prepare a sand mould of bend pipe.
3.	Inspect the grain fineness number of silica sand.
4.	Examine the moisture and clay content in the given moulding sand.
5.	Analyze the compression & shear strength of moulding sand specimen.
6.	Analyze the hardness of moulding sand specimen.
7.	Perform drawing operation.
8.	Prepare a vent rod using forging operations.
9.	Prepare a S-hook using forging operations.
10.	Prepare a lap joint & butt joint by arc welding process.
11.	Prepare a welded joint using Gas welding.
12.	Demonstrate of simple parts on a 3D printing Machine.
13.	Demonstrate of metal Casting process.
14.	Demonstration on ball milling, compaction and sintering.
15.	Demonstration of Chemical Vapor Deposition (CVD)

Reference Books:

1.	ElementsofWorkshopTechnologyVol-1Manufacturing Processes by S.K.Hajra Choudhury, A.K. Hajra Choudhury, Nirjhar Roy, MPP, Pvt. Ltd.
2.	Manufacturing Technology- Foundry, Forming and Welding by P.N. Rao, Tata McGraw Hill Publishing Company.
3.	J.R. Davis, Surface Engineering For Corrosion and Wear resistance, ASM International, 2001
4.	Ian Gibson, David W Rosen, Brent Stucker., Additive Manufacturing Technologies:

	3D Printing, Rapid Prototyping, and Direct Digital Manufacturing, 2/e, Springer, 2015.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620304	BS	--	--	3	1.5	50	50	100

CHEMISTRY FOR MECHANICAL ENGINEERS LAB

(For ME)

Course Objectives:

1	Demonstrate competency in using fundamental chemical principles and laboratory procedures to evaluate the chemical and physical properties of engineering-relevant samples.
2	To evaluate corrosion and electroplating of metals and introduce photocatalytic degradation for environmental remediation.

Course Outcomes: At the end of the course the student is able to

S.No	Outcome	Knowledge Level
1	**Quantify** the concentration, pH, and chemical composition of engineering samples using standard volumetric, spectrophotometric, and electroanalytical techniques.	K3
2	**Evaluate** the physicochemical properties of industrial water, electrolytes, and liquid fuels to assess their suitability for engineering applications.	K3,K4
3	**Synthesize** polymeric materials and functional nanomaterials using controlled laboratory procedures.	K3,K4
4	**Analyze** corrosion kinetics on metallic specimens and **determine** critical tribological properties (viscosity, cloud point) of lubricating oils under simulated operational conditions.	K3,K4

SYLLABUS

1	Determination of the concentration/strength of sulfuric acid in Pb-acid battery electrolyte
2	Determination of pH of water and soil samples using a pH meter.
3	Determination of the acid value of lubricating oil.
4	Determination of the saponification value of lubricating oil.
5	Determination of a strong acid against a strong base by using conductometer.
6	Preparation and study of a urea-formaldehyde/phenol-formaldehyde resin sample.
7	Determination of percentage of iron in cement sample by colorimeter.
8	Synthesis of nanomaterial by chemical precipitation method
9	Determination of the Cloud and Pour point of lubricating oil.
10	Determination of the corrosion rate of a mild-steel specimen by the weight-loss method

11	Determination of the viscosity of a polymer/nanoparticle-dispersion solution using an Ostwald viscometer.
12	Determination of Iron Content in Mohr's salt by Dichromate Titration
13	Estimation of Dissolved Oxygen by Winklers method.
14	Photo catalytic degradation of dye(Demo)
15	Determination of weak acid against a strong base by using potentiometer
Reference Books:	
1	Ahluwalia, V. K., & Dhingra, S. (2017). Comprehensive practical organic chemistry. Universities Press.
2	Vogel, A. I. (2012). Vogel's textbook of quantitative chemical analysis (6th ed.). Pearson Education
3	Virtual Labs, Ministry of Education, Government of India. https://vlab.co.in

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620305	ES	--	--	3	1.5	50	50	3 Hrs

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

(For ME)

Course Objectives: Student will learn

1	To verify Kirchhoff 's laws, know about safety measures used in electrical systems.
2	To determine power, power factor, electrical energy consumption, and solar power output under given operating conditions.
3	To conduct load test on single phase transformer, three phase alternator and 3-Phase Slip ring Induction Motor
4	About the characteristics and functioning of Zener diode and transistor, full wave rectifiers with and without filter
5	Verification of logic gates, Half Adder & Full Adder using Integrated Circuits,

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1	Verify KCL & KVL and determine power, power factor, energy in single phase circuit, solar power output at varying irradiance.	K3
2	Demonstrate the working of a 3-Phase Alternator, 1-Φ transformer, 3-Phase Slip ring Induction Motor under specified load conditions.	K3
3	Examine the operation of safety equipment under overload & short circuit conditions and DSO.	K4
4	Demonstrate the working of rectifiers, transistor as a switch and Zener voltage regulator.	K3
5	Verify the working of logic gates and half & full adders using integrated circuits.	K3

List of Experiments

1	Verification of KCL & KVL.
2	Measurement of Power and Power factor in single phase circuit
3	Measurement of Solar Power Output.
4	Measurement of Electrical Energy consumed by Domestic Electrical Appliances.
5	Load Test on a 1-Φ transformer
6	Load Test on 3-Phase Slip ring Induction Motor
7	OCC of a 3-Phase Alternator
8	Overload and Short circuit protection using Fuse / Miniature Circuit Breaker (MCB).
9	Measurement of amplitude and frequency using DSO.
10	Implementation of full wave rectifier with and without filter.
11	Transistor as a Switch
12	v-i characteristics of a Zener Diode and its application as voltage Regulator.
13	Verification of logic gates using Integrated Circuits (ICs).

14	Verification of Half Adder & Full Adder.
Reference Books:	
1	Principles of Electrical Engineering, V.K Mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2017.
2	Principles of Electronics Engineering, V.K Mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2017
3	Digital Logic and Computer Design, Morris Mano, Pearson India, 2016

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620021	MC	--	--	2	0	40	-	

NSS, SPORTS & YOGA

(Common to AIML, CSE, CSIT, IT, MECH)

Course Objectives:

1.	The objective of introducing this course is to impart discipline, character, fraternity, teamwork, social consciousness among the students and engaging them in selfless service.
2.	To make the students maintain their mental and physical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Understand the importance of discipline, character and service motto.	K2
2.	Explore human relationships by analyzing social problems.	K3
3.	Develop leadership skills and civic responsibilities.	K3
4.	Understand the importance of yoga and sports for physical fitness and sound health	K2
5.	Demonstrate health related fitness components.	K2

SYLLABUS

UNIT-I (10Hrs)	**Orientation** :General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance. **Activities:** i Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills ii Conducting orientations programs for the students –future plans-activities-releasing road map etc. iii Displaying success stories-motivational biopics- award winning movies on societal issues etc. iv Conducting talent show in singing patriotic songs-paintings- any other contribution.
UNIT-II (10 Hrs)	**Nature & Care** **Activities:** i Best out of waste competition. ii Poster and signs making competition to spread environmental awareness. iii Recycling and environmental pollution article writing competition. iv Organising Zero-waste day. v Digital Environmental awareness activity via various social media platforms.

	vi Virtual demonstration of different eco-friendly approaches for sustainable living. vii Write a summary on any book related to environmental issues.
UNIT-III (10 Hrs)	**Community Service** **Activities:** i Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media-authorities-experts-etc. i Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS, ii Conducting consumer Awareness. Explaining various legal provisions etc. iii Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education. iv Any other programmes in collaboration with local charities, NGOs etc.
UNIT-IV (10 Hrs)	Yoga Activities: Yoga practices- Asana, Surya Namaskar, Meditation.
UNIT-V (10 Hrs)	Physical Activities: Practicing general and specific warm up, Aerobics. Cardio respiratory fitness, Skipping and Running. Practice one major game and one individual sport(volley ball, basket ball).

Reference Books:	
1.	Nirmalya Kumar Sinha & Surajit Majumder, A Text Book of National Service Scheme Vol; I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2.	Red Book-National Cadet Corps-Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi
3.	Davis M. L. and Cornwell D. A., "Introduction to Environmental Engineering", McGraw Hill, New York 4/e 2008
4.	Masters G. M., Joseph K. and Nagendran R. "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi. 2/e 2007
5.	Ram Ahuja. Social Problems in India, Rawat Publications, New Delhi.
6.	The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc.2014
e-Resources	
1.	https://books.google.co.in/books/about/Social_Problems_in_India.html?id=jsvJoAEACAAJ&redir_esc=y
2.	https://openlibrary.org/verify_human?next=/books/OL32142863M/Social_problems_in_India%3Futm_source%3Dchatgpt.com